

	U.S.	JP	U.K.	GR	FR
Optimal Portfolio	0.45	0.24	0.16	0.11	0.04

This portfolio is heavily weighted towards the United States and Japan with weights of 45% and 24%, respectively. Note that by construction, this portfolio consists only of risky assets, so the portfolio weights sum to one. (The weights in this example are also fairly close to the market capitalization weights of these countries.) With a risk-free rate of 1%, the Sharpe ratio corresponding to this optimal portfolio is 0.669.

3.2. WITH A RISK-FREE ASSET

The addition of a risk-free asset expands the investor's opportunities considerably. Since there is only one period, the risk-free asset has no variance. Think of T-bills as an example of a security with a risk-free return. (There is some small default risk in T-bills, which you should ignore for now; I cover sovereign default risk in chapter 14.)

When there is a risk-free asset, the investor proceeds in two steps:

1. Find the best risky asset portfolio.

This is called the *mean-variance efficient* (MVE) portfolio, or tangency portfolio, and is the portfolio of risky assets that maximizes the Sharpe ratio.²¹

2. Mix the best risky asset portfolio with the risk-free asset.

This changes the efficient set from the frontier into a wider range of opportunities. The efficient set becomes a *capital allocation line* (CAL), as I explain below.

The procedure of first finding the best risky asset portfolio (the MVE) and then mixing it with the risk-free asset is called two-fund separation. It was originally developed by James Tobin (1958), who won the Nobel Prize in 1981. Given the limited computing power at the time, it was a huge breakthrough in optimal portfolio choice.

Let's first find the best risky asset portfolio, or MVE. Assume the risk-free rate is 1%. Figure 3.10 plots our now familiar mean-variance frontier for the G5 and marks the MVE with an asterisk. The dashed diagonal line that goes through the MVE is the CAL. (We encountered the CAL in the previous chapter.) The CAL starts at the risk-free rate, which is 1% in Figure 3.10 and is tangent to the mean-variance frontier. The tangency point is the MVE. The CAL is obtained by

²¹I have personally found this terminology a little confusing because "mean-variance efficient" portfolio sounds similar to "minimum variance" portfolio. Unfortunately this terminology is engrained, and I will also use it here.